

torch.quantized_batch_norm#

[https://pytorch.org/docs/stable/generated/torch.quantized_batch_norm.html#to](https://pytorch.org/docs/stable/generated/torch.quantized_batch_norm.html#torch.quantized_batch_norm)

Description:

Applies batch normalization on a 4D (NCHW) quantized tensor. input (Tensor) – quantized tensor weight (Tensor) – float tensor that corresponds to the gamma, size C bias (Tensor) – float tensor that corresponds to the beta, size C mean (Tensor) – float mean value in batch normalization, size C

Function Signature

```
torch.quantized_batch_norm(input, weight=None, bias=None, mean, var, eps, output_scale, output_zero_point) → Tensor#
```

Parameters

Returns

Return type

Returns:

Tensor

Code Examples

```
>>> qx = torch.quantize_per_tensor(torch.rand(2, 2, 2, 2), 1.5,
3, torch.quint8)
>>> torch.quantized_batch_norm(qx, torch.ones(2),
torch.zeros(2), torch.rand(2), torch.rand(2), 0.00001, 0.2, 2)
tensor([[[[-0.2000, -0.2000],
          [ 1.6000, -0.2000]],

          [[-0.4000, -0.4000],
          [-0.4000,  0.6000]]],

        [[[-0.2000, -0.2000],
          [-0.2000, -0.2000]],

          [[ 0.6000, -0.4000],
          [ 0.6000, -0.4000]]]], size=(2, 2, 2, 2),
dtype=torch.quint8,
  quantization_scheme=torch.per_tensor_affine, scale=0.2,
zero_point=2)
```

Detailed Documentation

Documentation

Applies batch normalization on a 4D (NCHW) quantized tensor.

input (Tensor) – quantized tensor

weight (Tensor) – float tensor that corresponds to the gamma, size C

bias (Tensor) – float tensor that corresponds to the beta, size C

mean (Tensor) – float mean value in batch normalization, size C

var (Tensor) – float tensor for variance, size C

eps (float) – a value added to the denominator for numerical stability.

output_scale (float) – output quantized tensor scale

output_zero_point (int) – output quantized tensor zero_point

A quantized tensor with batch normalization applied.